

## 9.7 Tabby

Tabby (<https://tabby.tabbyml.com>) is a self-hosted AI code wizard that consists of a server and an editor plug-in. The project is an exciting alternative to the combination of Ollama and Continue. Tabby is particularly suitable if a central server is supposed to provide code wizard services for multiple employees of a company or organization.

You can run the associated server either locally on your system or centrally in your company network. Tabby itself doesn't provide an LLM, but rather allows you to choose from a whole range of established LLMs specializing in development, such as StarCoder, Code Llama, or CodeGemma. The models are dimensioned in such a way that they can be run on a reasonably powerful system.

### 9.7.1 Installing the Tabby Server and the Plug-In

You typically operate Tabby in a container. The project provides a Docker image for this purpose so that you can get started straight away. If you have an NVIDIA graphics card, you can take advantage of the Compute Unified Device Architecture (CUDA). All you need to do is install the NVIDIA container toolkit. Then, start the container using the following command:

```
docker run -it --gpus all \  
  -p 8080:8080 -v $HOME/.tabby:/data \  
  tabbyml/tabby serve --model StarCoder-1B \  
  --chat-model Qwen2-1.5B-Instruct --device cuda
```

Use the following command for CPU-based execution:

```
docker run --entrypoint /opt/tabby/bin/tabby-cpu -it \  
  -p 8080:8080 -v $HOME/.tabby:/data \  
  tabbyml/tabby-cpu serve --model StarCoder-1B --device cpu
```

```
tabbyml/tabby serve --model StarCoder-1B \
--chat-model Qwen2-1.5B-Instruct
```

The `--model` and `--chat-model` options allow you to specify which model you want to use for code completion and chat, respectively.

As an alternative to running Tabby in a container, you can also install it natively on Windows, Linux, and macOS. The necessary steps can be found in the Tabby documentation at <https://tabby.tabbyml.com/docs/welcome>.

As soon as you run the Tabby server, you must create a new account, which you'll need later to connect to the server. You can access the Tabby web interface by default via `http://localhost:8080`. The first user you register here is automatically the admin user for the server.

You must invite all other users who are also supposed to use this server via the admin user settings. For this purpose, you need to enter the user's email address. As Tabby isn't connected to a mail server in its standard configuration, no email with the invitation code will be sent. However, you can copy the link to the invitation containing the code in the admin interface and send it manually to the person. This person must then complete their profile details and set a password.

The last step before you can use Tabby is to install the Tabby plug-in for your development environment. The plug-in is currently available for VS Code, IntelliJ, and Vi Improved (VIM)/NeoVIM.

## 9.7.2 Usage

As a concrete example, let's take a look at the Tabby integration in VS Code. After the installation, the plug-in prompts you to enter your token. This is available on the home page of the Tabby server. As soon as you've inserted and confirmed the token, the extension is ready for operation. You can use various supports in the extension:

- **Code completion**

While you're programming, Tabby suggests suitable completions. If the suggested code fits, you can accept it by pressing the `Tab` key. In your

daily work, Tabby often provides useful results here, which makes code completion one of Tabby's most important features.

- **Commands**

Via the command palette of the IDE, Tabby provides you with a series of commands such as `Tabby: Generate Docs` or `Tabby: Explain this`. To use such a command, you select a code and activate the corresponding command. The result is usually displayed in the chat view of the extension.

- **Code edits**

Code edits are a special type of Tabby command. You can use this command to have Tabby modify your source code. For example, you can select a function in a JavaScript file and give Tabby the instruction `Export this function`. The extension then inserts the appropriate `export` statement. In contrast to code completion, using code edits requires a little more patience, as the tool occasionally provides useless or empty answers. However, this feature provides great potential to catch up with the competition.

- **Chat**

In the chat, you can interact directly with Tabby's chat LLM and ask questions that will be answered directly.

Another exciting feature is the context in Tabby. You can specify a Git repository here, for example. Tabby processes the code from this repository and saves it in its internal index. The tool then uses this information for the extension's features such as code completion or contextual queries.